

Regularization

Mathematics of AI - RPTU Kaiserslautern

WiSe 25/26

Topics of the lecture

- Linear optimization with constraints
- Projection of a point onto an ellipse
- $Ax = b$ using L_2 (ridge) and L_1 (lasso) regularization

Exercises

Linear optimization with constraints

Let us consider the following situation: You own a store of furniture which produces only **tables** and **chairs**.

Each **chair** costs you 10 hrs and 20 units of wood. Each table costs you 5 hours and 30 units of wood. You can sell chairs and tables respectively for 80 and 60 euro. For next week you have 110 working hours, and 300 units of wood. What should you produce in order to obtain maximal possible profit?

Hint!

Find a way to represent the problem

Use `scipy.optimize.linprog`

Projection of a point onto an ellipse

Write a function that, given a point $p \in \mathbb{R}^2$, returns the projection onto the ellipse defined by the parameters a and b , which are given.

$$P(p) := \operatorname{argmin}_{x \in \Omega} \|x - p\|_2$$

$$\Omega := \left\{ x \in \mathbb{R}^2 : \left(\frac{x_1}{a} \right)^2 + \left(\frac{x_2}{b} \right)^2 = 1 \right\}.$$

If $p \in \Omega$, then $P(p) = p$

Hint!

This is similar to a problem you already saw.

L_1 and L_2 regularization

We want to solve again linear regression problem. But, with regularization.

- L_1 (lasso): $\min \|X\theta - y\|_2^2 + \varepsilon \|\theta\|_1$
- L_2 (ridge): $\min \|X\theta - y\|_2^2 + \varepsilon \|\theta\|_2^2$

Hint!

For quadratic programming you can use `cvxpy`. Solution of L_1 is *already provided*, if you want to give it a try, delete the cell :)

Remember how sometimes auxiliary variables can help.

Scikit-learn

In the last notebook, you can fit regression problems using L_1 , L_2 and combinations, but using the `scikit-learn` functions.

This package will be used in next hands-on, so take your time to understand how it works!